
Methods and Principles

What Is a Theory of Action?

Actions can be divided into two broad types, which relate differently to the behavior that instantiates them. The terms “action” and “behavior” are already the property of many different academic discourses and, since there already exists a huge literature on the philosophy of action, we need to clear some ground.¹

To understand what we mean by actions, one must first understand what we mean by society and what we mean by “form of life.” Societies differ: In British society in the 1990s we can go to the cinema, drive to work, play darts, supervise a student, take out a mortgage, and so on. If we were members of the Azande society studied by Evans-Pritchard, we could do none of these. We could, however, accuse someone of being a witch, prepare *benge*, consult the poison oracle, and invoke spirits; in current British society we can do none of these. The reason we could do some things in one society and not in another has little to do with what is legal and illegal, and everything to do with the differing social and conceptual structure of life in the two societies. It does not make sense to do the things in one society that do make sense in the other.² We are going to refer to those things one can intentionally do in a society, that get their sense from taking place in that society, as “actions.”

1. Anscombe (1979), Chisholm (1970), Danto (1973), Davidson (1982), and Goldman (1970).

2. Note that having read Evans-Pritchard's (1937) accounts of the Zande, we could “go through the motions” of doing some of these things in our own society, but we would not really be doing them.

Going to the cinema is an action; preparing benge is an action, and so forth.

We do other things that are not actions. Blinking is not an action. “Accidentally knocking over the table as a result of running through the house” is what philosophers would refer to as an action, but it is not the type of action that interests us. In the case of knocking over the table, our usage is different from the regular usage in the philosophical literature. In the case of blinking, our usage seems to be in accord with standard philosophical usage. A blink is something that happens to us, it is not brought about by something we do. A blink is merely a piece of *behavior*. But we will also use the word “behavior” in an extended sense; behavior includes all physical movements and therefore includes the physical movements humans use to execute the actions they intend.

Actions, Intentional Actions, and Formative Actions

Action Types and Tokens

The kind of actions that interest us are a subset of the kinds of actions that traditionally interest philosophers. We will be studying and classifying *types of intentional actions*.³ This emphasis upon types rather than *tokens* of intentional actions sets our project apart from the academic tradition, which deals with responsibility for actions and the logical form of action sentences. We are, then, studying and classifying action *types* like “(intentionally) greeting,” “(intentionally) swinging a golf club,” or “(intentionally) singing a song.” We are not studying action *tokens*, like Mary greeting John at 7:32 A.M. on Friday the 29th of October 1993, or like Fred singing “As Time Goes By” at 12:15 P.M. on the 5th of January 1992, except insofar as they usefully represent and illustrate action *types*.

Intentional Actions: Their Results and Their Consequences

Because of the stress upon types of *intentional* actions, we are able to avoid problems that have to do with the unintended consequences of intentional actions or with unsuccessfully executed intentions to act. Thus we are not concerned with actions such as “launching a torpedo

3. Our notion of action is similar to that of Carlos Moya’s (1990) “meaningful actions” (see chapter 4); see also Rubinstein (1977).

as a result of accidentally pressing the torpedo-launch button while intending to ring for the steward.” Nor are we concerned with the action of “missing the target” (except as it illuminates the meaning of trying to hit the target), and so forth. While for some analysts of action, the unsuccessful execution of intentions to act are themselves instances of types of actions (i.e., after all, “failing to open the door” can have many different instances), they are not, *ceteris paribus*, types of *intentional* actions. (Unless of course one fails intentionally.)

Generally speaking, the *result* of an action is the state of affairs that has to obtain for that action to have been carried out, whereas the *consequence* of an action is a further state of affairs that has been brought about by the attempt to carry out the action. The *result* of an action is conceptually linked to the intention while the consequences are contingently related only.⁴ If John opens a door and lets in cold air, we would say that the result of the action was the door being open, and the consequence was cold air entering the room—*unless* John intended to make the room cold by opening the door, in which case the cold air entering was also a result.

An example drawn from fiction shows nicely the difference between the results and consequences of actions. In John Le Carré’s (1963) novel of the Cold War, *The Spy Who Came in from the Cold*, Smiley, the controller of Leamas, a British spy, gives him the job of saving one East German spymaster and destroying another. The sensitive and refined Fiedler has been accused of being a traitor by the vicious Nazi, Mundt; Leamas is to save Fiedler and destroy Mundt. Leamas is told to assume the identity of a malcontent defector from the British Secret Service.

The ruse succeeds and Leamas is picked up by the East Germans and asked to give evidence at a secret trial. Leamas “betrays” the Nazi, telling the court that it is Mundt, not Fiedler, who has been working for the British. Mundt’s defense is that Leamas is a British plant, but the court does not believe Mundt. At this point Leamas believes he has achieved the intended *result* of his actions. He believes he has saved Fiedler, who has been helping the British, at the expense of Mundt, by concocting the tale of Mundt’s treason and having it believed by the East Germans.

4. See von Wright (1971).

Then Leamas's cover is blown and it becomes clear to the court that Mundt was right in accusing him of being planted by the British. Leamas's actions will not have the result he wanted; in fact they will have a *consequence* that is the opposite of what he intended. Fiedler is condemned and Mundt reinstated.

Later we learn that this was always Smiley's plan. Mundt, Nazi that he is, *is* a traitor to the East Germans and *is* working for the British; Fiedler is not. Smiley arranged for Leamas's cover to be blown at the crucial moment in order to save the vile Mundt at the expense of the honest Fiedler, who is in fact loyal to his own country, not to Britain.

Let us summarize: Leamas's intention was to save Fiedler at the expense of Mundt; that was the desired *result* of his actions. The unintended *consequence* of what he did in pursuit of that intention was to save Mundt at the expense of Fiedler. Note that for some philosophies of action, but not for ours, one would say that Leamas's action was to save Mundt. In our language, to save Mundt at the expense of Fiedler was Smiley's action and his alone; this was the intended *result* of Smiley's actions. Smiley used Leamas's quite different intention as a means of carrying out his own actions.⁵ The unintended *consequences* of Leamas's intention to act were the intended *result* of Smiley's actions.

Formative Actions

In one way the stories of Leamas and Smiley are poor examples of actions, as we use the term. To make the point, we will have to introduce a neologism. We differ from the majority of writers in the philosophy of action in that we are concerned with types of actions that constitute a community, a collectivity, or a form of life. Such actions make a society what it is and distinguish it from other societies.⁶ We call such actions

5. We will later develop the notion of an "action cascade." In an action cascade, one person's actions are included in the actions of another. Here Leamas's actions are included in the action cascade set in motion by Smiley.

6. The authors uncovered the differences in the extension of the term action that seem "natural" to philosophers and sociologists respectively only after many days of frustrating discussion. To a sociologist it seems perverse to use the same word "action" to refer not only to things we do intentionally that relate to the way we live, but also to accidents and unforeseeable consequences. A term with that

“formative actions.”⁷ Members of the same form of life share in a common net of concepts and actions. That is to say, they agree in their concepts because they share a realm of possible actions, and they agree in their actions because they share a common network of concepts. Actions and concepts are tightly intertwined because intentions are conceptual, and because concepts provide guidance for actions. This internal-essential link between concepts and actions notwithstanding, within a form of life some types of intentional actions are more formative (or constitutive) than others. Think, for example, of the form of life of Catholics in general, and the ritual during a mass in particular. It is not central to the ritual of a Catholic mass whether the priest sings along with the congregation. If, however, he was not to perform the action of praying for the transubstantiation, or if he did not lead the congregation in saying the Lord’s Prayer, most Catholics would feel either that the mass had not been performed *properly*, or had not been performed at all. We suggest calling these latter two actions, praying for transubstantiation and saying the Lord’s Prayer, *formative* of the Catholic form of life (or at least, of the form of life of the mass).

The borderline between formative and nonformative actions is fuzzy, and historically and culturally variable. After all, it is easy to imagine a sect within the Catholic Church that regards the *singing* rather than the *saying* of the Lord’s Prayer as absolutely essential for the successful performance of the mass ritual. That the borderline between formative

extension seems to the sociologist about as useful in the context of social science as a term that covers, say, all bicycles and all blue things. We discovered, eventually, that the sociologists’ “actions” corresponds to what many philosophers would call “types of intentional actions.”

7. We would have preferred to use the term “constitutive,” but this term is usually contrasted with “summary” or “strategic” as in constitutive and summary rules or constitutive and strategic rules. In our terminology, summary rules are as much constitutive of a form of life as are constitutive rules. Compare the rules of soccer as played by the Brazilian national team—a matter of graceful short passes—and by Wimbledon Football Club—a matter of getting the ball into the opponent’s penalty area as soon and as often as possible by long kicks upfield. In these cases only the summary rules are different, the constitutive rules being common to both styles of play. Nevertheless, the two forms of soccer life are different and are associated with a different range of possibilities of formative actions on the part of the players.

and nonformative actions can change does not mean, however, we are unable to draw a distinction in day-to-day practice. In fact, in many instances we are easily able to list a fair amount of formative action types belonging to different forms of life. Think for instance of the formative action types for different sports, for various games, for parliamentary sessions, for shopping, negotiating, lecturing in front of an academic audience, and so forth, and think of particular things that one might do in such settings that are not formative. Sometimes, of course, one innovates in such settings in the hope of changing a form of life and instituting new ways of going on.

Much of the philosophy of action has been developed in terms of examples that are not formative in our sense. Philosophers have concerned themselves mainly with examples of individual cases of assassinations, accidental killings, doing “this” while intending to do “that,” and so forth. This follows their preoccupation with questions of responsibility and causality, but the one-sided diet of examples has had the unfortunate effect of removing the whole issue of the interrelations between actions and forms of life from philosophers’ sight. From now on, unless otherwise indicated, we will use the term “actions” to mean “types of intentional actions that are formative.”

Actions, Institutions, and Fractals

Formative action types are social institutions, and they constitute social institutions. Let us sketch what we mean by an institution.⁸

What makes a metal disc “a coin,” or a piece of paper a token of value, that is, what makes such things instances of money? It is, of course, that a collective treats the items as money, that it talks of them as money, that it uses them as money, and that it sanctions and enforces this talk and this use. If the collective ceases to believe that the paper and coins are money, and that they have the value represented symbolically on their surfaces, these items cease to be money. This is not just a philosophical nicety but a practical problem, as inflationary spirals, and the precautions that governments take to prevent them, reveal.

8. For the performative model of social institutions, see also Anscombe (1974), Barnes (1983), Bloor (1997), Kusch (1997, 1998), Lagerspetz (1995), Searle (1995), and Tuomela (1995).

The institution of money is such that money ceases to exist if those who take part in the institution cease to act as though it exists. Both analysts and actors agree on that feature of money. In another kind of case, such as that concerning the notion of “mountain,” there is no question that the institution includes the idea that mountains would continue to exist even if the actors ceased to act as though they existed. We will use this distinction between “social kinds” and “natural kinds” in the analysis that follows.⁹ There has been much argument about the ways in which the two types of institution, those referring to social kinds and those referring to natural kinds, differ. For example, is the continued existence of mountains in the absence of actors’ beliefs about them to be treated as a feature only of the social institution of mountain or a feature of mountains themselves? But this argument is of no importance in this book.

Different societies have different social institutions; even when they have the “same” institution, they realize this “same” institution in different ways. For example, in the case of money, the guarantee might be reserves in gold, or the authority of a chief. Let us call this conventionality.

The mutual belief that sustains institutions is self-referring; the referent of the belief is the belief itself. On analysis, actors’ belief that a certain token is money turns out to be the belief that other actors also believe the same tokens to be money. The value of the money is nothing but mutual belief in the value of the money (or belief in the guarantee that lies behind it). It is also self-validating; it is the mutual beliefs that make the beliefs true.

Given the way such institutions are formed by mutually reinforcing beliefs, and given that this mutual reinforcement must be continual, it is not surprising that the form of social institutions is not fixed nor predetermined by any formula.¹⁰ Obviously institutions must be bounded

9. See also Barnes (1983), Bloor (1996, 1997), and Kusch (1997, 1998).

10. Barnes and Bloor refer to this feature of social institutions as “finitism.” The term contrasts with philosophical accounts in which the meaning of a term is taken to be its extension over an infinite set of instances. In Barnes and Bloor’s treatment, as in ours, meaning is based on a finite body of experience (Barnes, Bloor, and Henry 1996; Bloor 1997).

and rule following in some sense, or they would not be recognizable as institutions, but the way they are lived out is continually up for renegotiation—at least at the boundaries. Moving from context to context, actors continually reconstitute the institution through their decisions and actions. Such judgments are unavoidable, but they are underdetermined by past instances.

That judgments of how to act in new instances cannot be arbitrary is ensured by the fact that they must be collectively justifiable within the institution. In this sense, the “rule following” is not congruence with a formula, but mutual recognition that no mistake has been made.

The need for collective acceptance in social institutions makes the actions of members predictable; as we would say, it is one way in which action coordination is ensured. In this sense, formative action types are social institutions.

If we take forms of life to be larger sets of formative actions, they too can be understood to be institutions as described above. Thus there is a similarity of structure from “small” formative actions all the way up to whole forms of life. Think, for example, of the form of life of *sport* compared to the form of life of *war*. Each is driven by different ideas of what is appropriate to do. What counts as making a mistake is different in each. In sport it is not proper to kill to win, whereas it is proper in war. In sport (at least in some countries), it is considered better if the sides are reasonably fairly matched, whereas responsible leaders in a war will try to gain overwhelming superiority before committing their troops to battle. If a side of twelve-year-olds beats a side of five-year-olds by two hundred runs to two runs in a game of cricket, it will be said to have been a game organized by a fool, but when the ratio of casualties in the Gulf War was of the order of 10,000 to one, the allied generals were considered to have had matters well organized. The nature of these two social institutions—sport and war—is constituted and maintained by self-referential mutual beliefs.

Moving down to smaller examples, one finds, within the form of life of sport, the forms of life of individual sports. As social institutions these have again the same self-referential and self-validating “logic.” Sports differ not only in their encoded rules, but also in what is generally

accepted as proper ways of going on. Thus in golf (so we are told), it is expected that any infringement of the rules will be reported even if it was accidental, and even if the infringement was witnessed by no one but the golfer him- or herself. For example, if a player, unseen by anyone else, and gaining no advantage from it whatsoever, accidentally touches the sand in a bunker before hitting the ball, he or she is expected to report the misdemeanor and penalize him- or herself by the appropriate number of strokes. In modern professional soccer the opposite way of going on seems an integral part of the game: Soccer players will do all they can to hide their own infringements while pretending that others have fouled them, in order to have their opponents penalized unfairly.¹¹ It does not seem unreasonable to say that forms of life are embedded within one another as separate sports are embedded within sport as a whole.

It is not only forms of life that differ; roles within forms of life differ according to the types of acts that can sensibly be done within them. And again, self-referential mutual beliefs fix the nature of these roles. When we are supervising students we can suggest some reading, suggest a line of analysis, demand a piece of written work by a certain date, criticize

11. Golf is full of bizarre examples of players penalizing themselves. *The Observer* newspaper of Sunday, 7 November 1993, reports on page 5 of the sports section that the golfer Martin Poxon disqualified himself from a competition, and from automatic entry to all European tour events for the following season, by reporting a mistake that had no other conceivable consequences. During a competition played in very wet circumstances his ball ended up in a puddle on the putting green. Following what he thought was the correct procedure, he picked it up and dropped it away from the puddle but no nearer the hole; the resulting position happened to be off the green and on the "fairway." The ground being very wet, the rule in force that day was that a ball might be lifted from the fairway and placed in a "preferred lie" no nearer the hole and within six inches of its initial position. Poxon proceeded to "lift and place." The rule he had unwittingly broken, and that he did not discover he had broken until some time later, was that the ball should not have been dropped in the first place but should have been placed directly on the spot from which it was to be played. When compared to the antics of soccer players, the self-reporting of this mistake, with its devastating consequences, could not better express a difference between two forms of life. (We need not mention the bizarre consequences that befall a sport that tries to anticipate every eventuality in its rule book rather than relying on the good sense of officials and players. Golf is a good example of a misplaced attempt to eliminate polymorphic actions.)

the use of the passive voice, and so forth. We can't do any of this when we are, say, giving a public lecture at the university, though lecturing is just as much a part of our academic form of life as is supervising students.

Going still further down the scale, what we can do within roles also differs. When we are criticizing the use of the passive voice in a piece of written work, we might explain that it is often found in "scientific" literature to imply the lack of agency of the author. We might explain that it tends to take vitality out of the writing, we might point to particular sentences and show how they might be rewritten, and we might suggest the use of a style-checking software. Note, once more, that the Azande can do none of this, nor can we do it when we are giving a public lecture, nor can we do it when we are "demanding a piece of work from a graduate student by a certain date," even though the latter is still part of graduate supervision. Supervising a student is an action, criticizing use of the passive voice is an action, and suggesting the use of style-checking software is an action, though each of these actions is embedded within the previous one. And, again, what makes these actions the actions they are is that a group collectively takes these actions to be what they are, that is, the group constitutes them as these actions.

There is a mathematical idea that illustrates how, from actions all the way up to forms of life, we always can reidentify the same structure: the structure of self-reference, self-validation, conventionality, and lack of predetermination of what it is to follow a rule. This is the notion of the fractal. A fractal is a mathematical structure that retains its form no matter what the scale. The coast of Norway is a standard example: seen on the map, the coast of Norway has jagged inlets called fjords; each fjord, however, has its own jagged inlets so that the coast of a fjord has roughly the same form as the whole coast of Norway but on a smaller scale; each inlet within a fjord, if examined more closely, again has the same form, and so on down the scale.

The structure of forms of life and the actions that are embedded within them is like a fractal; their philosophical/sociological "morphology"—the self-referential, and self-validating structure of mutual belief—is preserved as the scale is reduced. Occurrences of fractal-like entities in nature do not have the purity of the mathematical structure; the similarity of form does not go on forever. In the case of Norway, by the time you get

down to grains of sand, the sequence is broken. The same is true of the form of life/action fractal. If we go a few steps further down the scale of what we do when we supervise a student, we can no longer be said to be acting. Thus in criticizing the use of the passive voice, we will be expelling air from our lungs—no longer an intentional action.

The fractal idea seems to capture the connection between the embedding social entity and the embedded social entity: In both, ways of thinking about things and ways of doing things are intimately connected. Thus there is a link between forms of life writ large (societies), forms of life within societies (war, sport), and forms of life writ smaller still (individual sports). As we go on down in scale we find ourselves no longer talking of forms of life, but of actions typically executed within them, and these again are linked in a fractal-like sequence. To understand actions as we use the term, it is important to see that the meaning of actions and forms of life are related all the way up and all the way down.¹²

A Technical Language for Actions

“When we do activity X, we do a lot of polymorphic actions and only a few mimeomorphic actions.” “Activity Y was once typically performed with polymorphic actions, but in modern times it is more usually done with mimeomorphic actions.” “In this group the activity consists largely of polymorphic actions, but in that group the same activity is constituted largely of mimeomorphic actions.” One problem in making this kind of claim sound convincing is that the normal labels that cut up our activities are taken from everyday talk, and the way everyday talk divides action does not fit tidily with our new technical categories. Suppose we say, as we will say, “swinging a golf club is a mimeomorphic action.” Someone

12. Collins and Pinch (1982) discussed these issues in connection with their studies of parapsychology. They felt that parapsychology and orthodox science were best understood as separate paradigms, or forms of life. Thus within one of these forms of life the results of an experiment would be seen as signifying one thing, whereas in the other form of life, the same experimental result might be seen as signifying its opposite. There were, however, high-level elements in common, such as the notion of experiment itself.

might remark to us that they swung a golf club only yesterday and did it after the fashion of a polymorphic action. That we open ourselves to this kind of response is a result of our trying to express ourselves briefly. What we should have said is: "Since the 1930s when someone swings a golf club, and that person is trying to swing the golf club according to the method recommended in the contemporary coaching manuals that help to make up the form of life of modern golf, then they are doing a mimeomorphic action." Actually, that is how most people do try to swing golf clubs today, and we allow ourselves the luxury of abbreviating the exact long formulation by the shorter "swinging a golf club is . . . , etc." The usage is sociological; it expresses something about how things are generally done in this society and how they are expected to be done.

We claim that if we explained what we meant about different types of action to actors in any particular sphere of action, they would be able to see what we were getting at within their practice, even though the concepts we bring to bear are "technical concepts." Technical concepts are ways of describing the world that are not in widespread use. Even if the occasions on which the two types of golf swing were practiced in real life were horribly intermingled, a golf player would still be able to see that there are two types. Nevertheless, seeing what is going on in terms of types of action can sometimes be hard because we have such a fixed idea of how we do things that we are often unable to reflect properly upon them.

Individuals, Society, and "Internal States"

We have said that we are concerned not with particular actions done by particular individuals on particular occasions, but with action types that are formative of societies. This does not mean we can escape entirely from what goes on in peoples' heads. Actions, as we use the term, are tied up with intentions, and intentions are internal states. It is impossible to be sure about internal states. When it is important to identify individual internal states, as in assignment of responsibility for crimes, huge and costly institutions are set up to investigate them. Despite these purpose-built institutions, intentions are sometimes misidentified. But attributing an internal state to an individual is quite different from investigating the internal states that are plausible features of a form of life. That we can

never be sure whether we have uncovered the motives of a murderer is not the same as saying that we can never be sure whether or not basely motivated murders take place in our society. It is easy for us to understand how this *type* of action could take place, even if we cannot be sure that we have correctly identified any one of its instances, or tokens. We can understand that basely motivated murders could take place because we are participants in the society and because we know that murder is a feature of it.

Suppose that every night between twelve and one we engage in certain movements that we think of as doing a rain dance, with the intention of making it rain the following day. Sometimes we believe we are successful, and sometimes not. The idea of a rain dance has to be familiar to us if we are to think of ourselves as rain dancing, but at the same time we are sure that the performance of dances meant to bring rain is not part of any of the forms of life pertaining to British society. What sort of “sureness” is this? It is the sureness of the enculturated participant in the society. Getting to know the kinds of actions that might reasonably and regularly be done in a society is a lot easier than identifying whether some action was actually done by an individual. It is this kind of sureness that allows sociologists to discuss, say, the differences between Zande and Western society.¹³

Though getting to know individual motives is the province of priests, psychiatrists, and judges and juries rather than of sociologists or anthropologists, some part of coming to participate in a society involves talking to individuals. Asking individuals to discuss their intentions is one of the ways in which we might develop our understanding of how aspects of society work in general. In these cases the concern with individuals is the way they represent general patterns of action. Individuals are indices of the nature of the underlying collectivities.

Intention and Consciousness

What we have said so far might be misconstrued as meaning that an intentional action is always something that follows on a consciously

13. It would be a problem not only for interpretative sociology but even for quantitative research; even questionnaire design turns on shared understandings of what can be done in the way of asking and answering questions.

formulated intention. But an action can be an intentional action without the actor having to be aware of the intention from moment to moment. Imagine you are driving your car to work and you are changing gears while thinking about something else entirely. Then, for a moment, your attention strays back to the car driving during a sequence of gear changes. We do not want to say that the first set of gear changes were not actions while the second set were. It cannot be that what counts as an action switches around in this way depending on exactly what the actor is thinking about; that would be an unsatisfactory notion of action.

Sometimes the intentional nature of actions can be made apparent if actors are disturbed and asked what they are doing.¹⁴ If, during your car-driving reverie, an irritating sociologist were to ask you what you were doing and why, you would be able to reply; this is one way of providing a warrant for saying the action was intentional. Nevertheless, were we to disturb a car driver who denied all knowledge of changing gears, we would still not doubt that the gear changing was an action.

As we will argue later, we do not require that actors are aware of what they are doing when they are woken from their reveries; we require only that *some* actors know what they are doing when woken from reveries; this is enough to establish the existence of a *type* of action.

What Is “The Same” Behavior?

Our theory rests upon the relationship between action and behavior, and therefore we have to discuss action and behavior as distinct from one another. One of the sorts of things we will want to say is that there are certain kinds of actions that are distinguished by our desire to perform them with the same behavior on every occasion. But to say things like this presupposes we know what we mean the “the same.” We have to recognize that “the same,” even when it is a matter of physical movements, is not the same for everyone. We will return to the “sociology of the same” in a later chapter. To introduce an example that we will return to later, one golf swing may feel the same as another golf swing to a golfer, especially if both result in the ball traveling the same 250 yards on the same flight path, but these two golf swings would look different

14. See von Wright (1971) and Dennett (1991).

to the owner of the hidden dog that is hit by the second ball landing one yard from where the first landed. An engineer watching the two swings on videotape or making measurements with a micrometer would also see them as different.

Who might be included among those whose groupings of sameness must be taken into account as we develop our theory? There is the actor, there are observers of the action who do know what action is being essayed, and there are observers who are not in a position to know what action is being essayed, or who, like the dog owner or engineer, have different interests than those of the actor. There is a tension in sports commentaries, both professional and amateur, caused by the differences in perspective. When a golfer hits a shot of, say, 200 yards that finishes within a foot of the hole, commentators and playing partners will call it a “great shot.” However, a shot of this sort is no better in terms of human accomplishment than a “good shot” that finishes six feet from the hole; at that range the difference between one foot and six feet is chance alone.

Actors and knowledgeable observers might refer to groups of *physical movements* as “the same” by grouping them according to the sameness of the intentional action. Thus there is a class of soccer strikes on goal that could be taken as exhibiting the same behavior irrespective of the final position of the ball, inside or outside the net.

Let us switch examples and consider the action of uttering a certain word—for example, “you.” Utterers and knowledgeable observers could agree that all mouth movements associated with saying “you” were “the same movements”—the movements required to say “you.” But someone who did not understand the word “you” would be most unlikely to think that all the mouth movements associated with the myriad ways of pronouncing “you” among different people should be classed as the same; they would have no reason to see them as the same.

The class of actions we call mimeomorphic actions is the class where the kind of trouble discussed in the last paragraph need *not* occur. It is worth spelling this out:

Mimeomorphic actions are actions where exact reproduction of the *behavior* by someone who *did not understand the action* would

always appear to reproduce the *action* to someone who *did understand the action*.¹⁵

One way this kind of act can be facilitated is by the actors always *trying* to reproduce the behavior that instantiates the act in as near as possible the same physical way each time it is essayed. Then imitators need learn to reproduce only a set of physical movements to mimic the action. Success is not guaranteed, but it is a good start.

The notion of sameness is tied to the notion of digitization—that is, making sure that there is an area of tolerance around each entity that is to count as the same.¹⁶ It is the tolerance that makes it possible for mimeomorphic actions to be imitated by an observer because the mimicry can be inexact. Life can be made easier for uncultured imitators by making the movements “coarse grained” on the human scale. It is easier to detect arm movements than mouth movements, and if arm movements are themselves arranged into a course-grained system of categories, it is still easier. That is how semaphor works. First, words are digitized by the use of an alphabet, then, the letters are represented by course-grained arm movements.

Using semaphor, we can signal the word “you” without having any idea that we are doing it. To make “you” using semaphor we *could* simply intend the bodily movements without thinking about the word. What is more, an observer who did not speak English could mimic the movements of the arms and thus pass on the letters intended to someone who could speak English, and who would then understand the word. The coarse-grained digital system of semaphor consists of “left arm” or “right arm,” with eight positions for each arm (up, down, straight out left, straight out right, and the four in-between points). In semaphor, “you” looks like:

\ _	_ \	\ /
Y	O	U

Prior to recognizing a semaphor signal, one must recognize the categories of arm movement; how one learns to see this is a problem, but

15. And this is how the “Chinese Room” (e.g., Searle 1980) is supposed to work!

16. See Haugeland (1985) and Collins (1990).

the only point we want to make here is that it is possible. It is not possible to do this with mouth movements.¹⁷

Natural and Social Kinds

We can use the difference between social kinds and natural kinds to clarify the distinction between polymorphic actions, mimeomorphic actions, and (mere) behaviors. Both natural and social kind terms are social institutions in the sense that they have a self-referential component. But, as we have said, the self-referential component works differently in the two cases. In the case of money, there is nothing but self-reference—the reference is, as it were, “exhausted” by the self-reference. In the case of mountains, the term refers away from itself toward the physical world which is treated (within the institution at least), as existing independently of the reference. The continuing self-referential component of “mountain” consists of the criteria for classifying individuals as mountains. Nothing can be a model for anything in and of itself; a model of a mountain is what is collectively taken to be a model of a mountain, and that which is “Everest” is what is collectively taken to be “Everest.”

Now let us put this distinction to work on the three things that humans do. Mere behaviors are recognized by means of natural kind terms—that is, mere behaviors are natural kinds.

Actions are social kinds; that is, actions are not just social institutions, they are social institutions in which self-reference exhausts the reference. Actions like “greeting,” or “saying the Lord’s Prayer,” or “swinging a golf club” are like “money”; they exist because they are collectively constituted. Something is a greeting because it is collectively taken for a greeting; something is money because it is collectively taken to be money; something is a golf swing because it is collectively taken to be a golf swing. Actions are social kinds because communities are able to recognize their behavioral instantiations.

Polymorphic actions fit the above definition uniquely; a polymorphic action can never be specified by listing the behaviors in terms of which

17. It is, of course, just as easy to identify the printed letters “Y,” “O,” and “U” as the semaphor positions. The crucial comparison is between lip movements and arm movements.

it could be carried out. Thus one cannot specify a polymorphic action by means of natural kind terms.

A piece of behavior like a blink or other reflex is at the other pole of the dichotomy. A piece of behavior is part of the physical universe, and for certain purposes we can take such behaviors to be, like mountains, independent of our classifying activities.

Mimeomorphic actions are especially interesting because they have qualities belonging to both poles of the institution dichotomy. Because mimeomorphic actions are associated with fixed sets of behaviors, they can be identified in terms of natural kind terms. But a mimeomorphic action remains an action and can be looked upon, and identified as, a social kind; so, as an action (as opposed to a set of behaviors), its reference is exhausted by self-reference. Its behavioral instantiations are recognized by a collective, and this recognition is what makes this group of behaviors correspond to *this* rather than *that* mimeomorphic action.

Thus a Nick Faldo-type swing can be a natural kind term (as described in bodily movement terms by Nick Faldo's coach), and can be used to pick out a certain type of behavior from the other behaviors in the world around us. Note that picking out behaviors in this way is relative to cultures in the way that identifying mountains is, but not in the way that identifying polymorphic actions is. One does not have to be a member of the behavior's culture to identify the set of behaviors corresponding to a mimeomorphic action, but one does have to be a member of the culture to know which mimeomorphic action it corresponds to.

To sum up: The distinction between action and behavior maps onto the distinction between natural and social kinds. Polymorphic actions can be identified only by means of social kind terms. Mere behavior is solely a natural kind. Mimeomorphic actions can be identified by social kind terms, by natural kind terms, or by both. Behaviors carried out by machines that correspond to mimeomorphic actions can only be identified by means of natural kind terms, but may appear to constitute social kind terms.

Different cultures might have different taxonomies of human movements: For one culture a Nick Faldo-type swing behavior might be parsed into one sequence of movement descriptions, while for another culture a rather different set of movement categories might be used for capturing

the same movement (just as some cultures might divide the world of precious and nonprecious stones differently than others do).

Nevertheless, it is characteristic of what we mean by behavior that different cultures can employ their different taxonomic resources for achieving the same effect: the repetition of the same (sequence of) behaviors. What we have just claimed about cultures also holds for machines, of course. A machine might mimic a human behavior—say a golf swing—without using any of the entrenched ways of parsing human behavior. It is able to repeat the behavior, although its (stored) description of the behavior is unlike any of the human descriptions.

Natural and Social Science

The difference between similarities of action and similarities of behavior can now be used to redescribe the difference between a natural and a social science. Imagine a group of “scientists” watching a group of “Saturnians.” Saturnians do nothing all day except signal to each other in semaphore about the weather; the scientists do not know that. Saturnians know only 100 messages about the weather.

The scientists would have some chance of spotting regularities in the types of Saturnian behavior described above. They might first think that the Saturnians were “waving their upper limbs about.” Then, after further study, they might note that the upper limb movements could be grouped into a eight different “positions” (given that angles between positions were rarely held for any length of time). Then some analyst (or computer program), might notice that the movements could be grouped into a limited set of sequences. Finally, some genius might notice that the sequences were correlated with the weather. The observers would thus have developed an observationally based science of Saturnian upper limb movements. It would include theories such as “the following sequence of upper limb movements is correlated with the appearance of the sun,” and “such-and-such a sequence of upper limb movements is correlated with a hydrofluoric acid shower,” and so forth. The observers will come to be able to predict what upper limb movement will come next by watching the horizon. Such a science has at least a chance of developing. It is a science of behavior made possible because the behavior of Saturnians is systematically related to their intended actions—making remarks

about the weather. This, then, *pace* Weber,¹⁸ is a science of intentional action made possible because, from the outside, it is indistinguishable from behavior. It may well be, of course, that the science does not develop for all the reasons that any science does not develop. Perhaps Saturnian Morse code is transmitted by tiny finger movements that earthly scientists do not have the visual acuity to detect; perhaps the Saturnians' movements are too rapid to see. All we are saying is that *one* reason for the science not to develop—the lack of order in behavior when it is motivated by polymorphic action—is not a problem.

Suppose, on the other hand, the Saturnians, not wishing to be “understood” by all and sundry, incorporated a complex code between their semaphoric movements and the symbols of their language that the movements represented. Let us say that those in the know learned this code every night from a randomly generated and secretly distributed “one-time” code sheet. The behavioral science of Saturnians would not then develop; the observers would never discover the ordering and the correlations of the arm movements.¹⁹ Where any action is concerned, observers who do not understand the “code” that links behavior to action will, likewise, be unable to develop a predictive, observational science of the *behavior* found in the societies they observe.

In the case of polymorphic action, the “code” that links intention and behavior is complicated, cannot be set out, and can only be known through participation in the society being studied. There is nothing exotic about this claim. There is no more to it in principle than pointing out that one simply cannot recognize all the mouth movements associated with the utterance “you” unless one can recognize the sound “you” as a result of one’s understanding of what is being said. This claim is given enormously strong empirical support by research into speech recognition by computers.

18. Gerth and Mills (1948).

19. What right do sociologists of science have to provide such a simplistic picture of science making? Even with nice ordered arm movements, the science might not develop because observation is not exact, because mistakes will be made, and because the scientists will argue; data must still be separated from noise. What is more, even if Saturnian arm movements were not systematically related to the weather, it is possible that the science *would* develop, so long as the scientists

The game of “Chinese whispers” gives a good indication of the differences discussed here. Imagine that the in-between people in a Chinese whispers chain contained only individuals who did not understand the language. We tried a small experiment to illustrate the point. Four graduate students at the University of Bath participated. None of the students knew semaphor, but they did know that in semaphor each arm was allowed to take one of eight positions. They were told a sequence of arm positions to learn and teach to a third party. They did not know what the arm positions signified. Nevertheless, the three letters Y-O-U were successfully transmitted through the chain.²⁰

We then tried the same experiment with the French word “toi” using lip movements instead of arm movements. This time, when the final person in the chain was told that the word to be identified was a French word, she identified it incorrectly as “oui.” The next experiment is to try Chinese whispers using skilled lip-readers working in a language they do not understand.

Sciences fail for two reasons (among many others). They fail because the things they try to correlate are randomly or chaotically associated in nature; long-range weather forecasting fails for this reason. Or they fail because the scientists do not understand the code. In the case of the social sciences, the necessary “code” cannot be understood except through participation.

Now we come to a crucial point. In the case of mimeomorphic actions, we could, in principle, have a purely observational, predictive science of behavior. It might be a lot easier and quicker to develop an understanding first, but in principle an outside observer could work out what counted as “the same” behaviors in a society and would have a chance of being right. Outsiders’ and insiders’ categories of sameness could match. Social science would then fulfill B. F. Skinner’s dream and be a purely

were ingenious enough. The statements made here rest on a rather naive view of science as something that makes theory correspond with regularity found in nature, and we know it is not really like this. But what other resource have we if we want to speak of the possibility and impossibility of a science? The debate about the nature of the social sciences and the nature of methodology turns on this kind of thinking.

20. We did have some initial problems with mirror-image reversals.

“behavioral” science. In the case of polymorphic action, this is impossible.²¹

Two Kinds of Order

It is easy to overlook the distinction that has been made between orderly *behavior*—for which we can have a purely observational science—and orderly *action*—which requires the social scientist to gain a participant’s comprehension if it is to be understood. It is easy to overlook this difference because we all have an enormous amount of low-level expertise in social affairs, and therefore we do not notice the extent to which we are skilled understanders even before we essay social scientific research.²²

The two kinds of actions we talk about, polymorphic actions and mimeomorphic actions, give rise to *two kinds of order* in social life. Ordinary social life is full of order, but it is ordered polymorphic action, mutually understood through common enculturation. This is not the same as ordered behavior.

The matter is still more confusing, because at higher levels, orderly action can be organized within orderly behavior.²³ To give an example, under certain restaurant management regimes the manager of a restaurant might worry only about how many meals a waiter has served; as far as the manager is concerned, each meal served is like every other meal served. An observer watching the manager (even if not understanding), might spot the orderliness of the manager’s behavior. Suppose the waiter hands in a “chit” every time a meal is served, and the manager gives the waiter a ten-dollar bill (as wages) every time ten meals are served. This

21. Note that it is not the case that one can *always* identify even a mimeomorphic action merely by observing behavior. The Saturnian case works because the scientist observers are making statistical generalizations. But a single action may fail; then the resulting behavior will not correspond to the intended action even though the actor intends it to correspond. (Perhaps one of the Saturnians is wearing a tight vest and cannot move its arms freely enough to execute its intentions.) Also, a single behavior might correspond to many actions.

22. This point will be dealt with again in a later chapter under the heading of “RAT.”

23. The idea for the following example comes from Phil Agre’s (1994) paper “The Grammar of Action.” Agre is not responsible for the “types of action” element in the discussion.

correlation—between chits going one way and green pieces of paper going the other—could be spotted without any understanding of restaurants. But to understand what counted as the serving of a meal would be another matter altogether.

The distinction between polymorphic actions and mimeomorphic actions can be seen, as has been remarked, as a distinction between who (in principle) is able to see the order in it; who is able to make reasonable judgments of sameness about the behaviors they see. In the case of polymorphic action, only the enculturated can see sameness properly; in the case of mimeomorphic actions, nonenculturated observers might see it too. As it happens, there is hardly any situation on earth in which observers watching a society might really see regularities of behavior without any understanding, because on earth mimeomorphic actions are nearly always intimately mixed with polymorphic actions, and this would confound the sort of science of behavior that works on Saturn. It is only in strange and unusual circumstances that a science of behavior could really work.²⁴

24. Surprisingly, many economists believe all action is of this type.